

Digital Financial Inclusion and Fraud Exposure in Italy

Internet Access, Online Financial Adoption, and Security Awareness among Italian Adults

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Abstract

This study examines digital financial inclusion in Italy as a sequential process moving from Internet access to online financial adoption, frequent use, exposure to adverse events, and protection through security awareness. The analysis uses the IACOFI 2023 survey. We first investigate Internet exclusion through weighted descriptive statistics, an age–education heatmap, corrected Cramer’s V , robust weighted logistic regression, odds ratios, and average adjusted predictions. Thanks to these results we concentrate the analysis to connected adults and study the financial onboarding ,done completely online, through five QP8 items and the frequency of seven different digital financial activities using QP9. We then evaluate the predictors of reported digital financial fraud exposure using a weighted binomial logistic model with HC1 robust standard errors. Finally we tested whether a ten-item, direction-corrected security-awareness index is associated with lower incident probability after accounting for digital-finance intensity and socio-demographic controls.

Keywords: Digital Inclusion, Internet Access, Digital Financial Services, Online Adoption, Fraud Exposure, Financial Incidents, Security Awareness, Logistic Regression.

Dataset Selected: Indagine sulle Competenze Finanziarie e le Competenze Digitali Finanziarie degli Adulti (IACOFI 2023), Banca d’Italia.

AI Tool Declaration: Generative AI was used for the technical refinement of the programming pipeline and LaTeX formatting. All the variables chosen for the analysis, the whole process for the construction and the interpretation of the analysis remain entirely the the original work of the authors.

Reproducibility: [GitHub repository](#).

1 Research Questions

1.1 Overall Framework

The project studies digital financial inclusion as a sequence of connected but distinct stages. Access is the necessary starting condition, but it does not guarantee adoption, frequent use, or safe participation. Once individuals use digital financial channels, they may face greater exposure to phishing, scams, unauthorised card use, or unrecognised transactions. Security knowledge and behaviour may then reduce the probability that exposure becomes an adverse incident.

The complete conceptual pathway is:

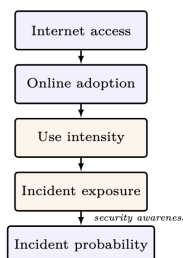


Figure 1: The digital financial inclusion pathway underlying the four research questions.

The arrows represent an analytical sequence rather than a deterministic funnel. A connected adult may use online banking without having opened a product entirely online, while a reported incident reflects not only underlying exposure but also recognition, recall, and willingness to report.

1.2 Statement of the Research Questions

RQ1a — Internet Access.

Which demographic groups are more likely to be excluded from Internet access in Italy?

The first part identifies the groups unable to reach the starting point of digital financial inclusion. It considers age, education, gender, employment status, macro-region, and urbanisation. Household income is included only in a robustness specification because non-disclosure is substantial and may not be random.

RQ1b — Digital Financial Adoption and Use.

Among adults with Internet access, how are age and education associated with fully online financial onboarding and the frequency of digital financial activities?

This part separates connectivity from effective use. Fully online onboarding and current activity intensity are treated as related but non-equivalent outcomes.

RQ2 — Activity and Reported Fraud Exposure.

How are Internet-use frequency, perceived digital/financial knowledge, the number of financial tools owned, age, and education associated with the probability of reporting a digital financial fraud attempt or incident?

It investigates whether operational exposure and knowledge are more closely associated with the reported outcome than demographic characteristics alone.

RQ3 — Protective Role of Security Awareness.

At the same level of digital financial activity, is greater security awareness associated with a lower probability of reporting a financial incident?

We tested whether security-relevant knowledge, attitudes, and behaviours reduce incident probability once exposure is held constant, and whether this protection varies across levels of use intensity.

1.3 Motivation and Stakeholders

Digitalisation creates both opportunities and risks. Lack of Internet access prevents independent use of online banking, payments, insurance platforms, and investment services. Among connected adults, difficult onboarding, low trust, limited familiarity, or preference for traditional channels may still prevent adoption. Among active users, the number and frequency of digital interactions create more opportunities for fraudulent contact. The policy problem therefore changes across the pathway.

The analysis is relevant to Banca d'Italia, commercial banks, FinTech firms, consumer associations, public administrations, and organisations involved in digital inclusion. Excluded adults require assisted access and viable non-digital channels. Connected non-adopters require simpler and more accessible onboarding. Intensive users require stronger preventive communication, safer interface design, and security support proportionate to their exposure.

1.4 Hypotheses and Empirical Expectations

- **H1:** older age and lower education are associated with a higher probability of lacking Internet access.
- **H2:** among connected adults, online financial adoption differs non-linearly by age and is higher among more educated respondents.
- **H3:** frequent digital financial activity is more common among younger or middle-aged respondents and among people with higher education.
- **H4:** greater Internet-use frequency is associated with a higher probability of reporting a digital financial fraud attempt or incident

All expectations concern statistical association. The cross-sectional design does not permit causal or formal mediation claims.

2 The Data

2.1 Data Description

The analysis is based on the IACOFI 2023 survey [1], administered to a representative sample of Italian adults by Banca d'Italia. The raw dataset contains 4,862 observations and 219 variables covering socio-demographic characteristics, Internet access, general digital activities, financial products, fully online onboarding, frequency of

online financial operations, knowledge, attitudes, security practices, adverse events, and survey weights.

The population weight `wght` is used for population-oriented descriptive percentages and pseudo-weighted regression models. The available materials do not include complete information on strata and primary sampling units; consequently, the inferential models are not full complex-survey estimators.

2.2 Preprocessing and Sample Construction

Internet access. The variable `internet_access` retains only valid 0/1 responses from `qd14`. The complementary outcome `no_internet` is defined as $1 - \text{internet_access}$, so that a positive event represents exclusion. Of 4,862 respondents, 4,427 report access, 415 report no access, and 20 have a missing or invalid response.

Age. Exact age from `qd7` is retained within a plausible adult range. Missing exact ages are recovered using `qd7_a`. RQ1 uses six bands. RQ2 collapses age into 18–39, 40–59, and 60+. These different groupings are preserved because they reflect the supplied analyses.

Education. The access analysis uses university, upper secondary, lower secondary, and primary or less. The adoption analysis combines the two lowest groups to reduce sparse cells. RQ2 uses High, Medium, and Low categories.

Special response codes. Values such as -97 , -98 , -99 , and -999 represent refusal, non-response, or routing-related cases. They are treated as invalid or missing rather than as substantive negative answers.

Connected subsample. RQ1b and RQ3 are restricted to the 4,427 respondents with Internet access. People without access are not classified as ordinary non-adopters because the relevant digital-service batteries are not meaningful or are not administered to them.

2.3 Operationalisation of RQ1 Indicators

The online-adoption indicator is:

$$A_i = \begin{cases} 1, & \text{at least one valid QP8 item equals 1,} \\ 0, & \text{all observed QP8 items equal 0,} \\ \text{NA,} & \text{otherwise.} \end{cases} \quad (1)$$

A count, `qp8_number_services`, records the number of positive QP8 items from 0 to 5.

QP9 responses use a four-point frequency scale: 1 = Never, 2 = Sometimes, 3 = Often, 4 = Very often. The binary indicator `any_frequent_qp9` equals 1 if at least one valid activity is reported as Often or Very often, and 0 when all observed activities are Never or Sometimes. The respondent-level mean is calculated only when at least four of the seven QP9 items are valid.

QP8 and QP9 measure different concepts. QP8 asks if a person has ever done any onboarding operation while QP9 concerns current behaviour (considering last 12 months). A respondent may therefore be a frequent digital user without having completed the original onboarding entirely online.

Table 1: Variables used across the three research questions.

Dimension	Variables	Operational interpretation
Internet access	qd14	Binary availability of Internet access; valid values are 0 and 1.
Age	qd7, qd7_a	Exact age where available, with categorical fallback; grouped into 18–29, 30–39, 40–49, 50–59, 60–69, and 70+ for RQ1.
Education	qd9	Four categories in the access analysis; three categories in the adoption analysis.
Gender	qd1	Woman or man in the access and RQ3 control models.
Employment status	qd10	Detailed categories in the access model and controls in RQ3.
Macro-region	qd2	North-West, North-East, Centre, South, and Islands.
Urbanisation	qd3	Rural area, small municipality, medium urban centre, city, or large city.
Income	qd13	Declared income groups plus non-declaration; used only for robustness in RQ1a.
Online onboarding	qp8_1–qp8_5	Account, card, insurance, credit, or investment platform opened/subscribed entirely online.
Digital financial activity	qp9_1, qp9_3–qp9_7, qp9_10	Frequency of balance checks, bill payments, online purchases, transfers, product management, smartphone payments, and robo-advisor use.
Reported fraud exposure	QP10 family	Binary RQ2 outcome indicating a reported or recognised digital financial fraud attempt or incident.
Internet-use frequency	internet_freq	Engineered RQ2 intensity measure based on digital-banking and e-commerce activity.
Perceived knowledge	internet_knowledge	Engineered digital/financial knowledge measure from the QA/QF sections in the supplied RQ2 analysis.
Financial tools	num_financial_tools	Count of financial instruments owned, derived from QP1.
Financial incident	qp10_1–qp10_4	At least one investment scam, phishing event, unauthorised card use, or unrecognised transaction in RQ3.
Security awareness	QK7, QS2, QS3, QS4	Direction-corrected and standardised index combining ten security-relevant knowledge, behaviour, and attitude items.
Population weighting	wght	Positive survey weight used in weighted percentages and pseudo-weighted models.

2.4 RQ2 Fraud-Exposure Variables

The RQ2 dependent variable is derived from the QP10 family and indicates that a respondent experienced, identified, or reported a digital financial fraud attempt or incident according to the coding in the supplied analysis. The predictors are `internet_freq`, `internet_knowledge`, `num_financial_tools`, age group, and education group.

The source document reports the conceptual origin and regression output, but it does not fully document every coding threshold, missing-data rule, or the final complete-case sample size. The odds ratios for continuous engineered predictors therefore refer to a one-unit increase on the implemented scale; the final repository should report the exact scale ranges.

2.5 RQ3 Security-Awareness Variables

The RQ3 analysis uses three principal indicators.

Digital-finance intensity. The intensity score is the row-wise mean of seven QP9 frequency items, each coded from 1 to 4, requiring at least four valid items. It is available for 4,240 respondents, with mean 2.07 and standard deviation 0.67.

Security-awareness index. Ten items from QK7, QS2, QS3, and QS4 measure security-relevant knowledge, password behaviour, data-sharing practices, provider verification, and safe-browsing attitudes. Each item is recoded so that higher values consistently represent greater awareness, standardised to a common z -scale, and averaged.

The direction corrections applied to the individual items are summarised in Table 2. At least six valid items are required. The index is available for 3,889 respondents and is approximately centred at zero by construction (mean 0.00, standard deviation 0.41)..

Financial incident. The binary outcome `financial_incident` equals 1 when the respondent reports at least one of four adverse events: investment scam, phishing, unauthorised card use, or unrecognised transaction. It equals 0 only when all four items are validly observed as negative. Among 4,189 classifiable connected respondents, 678 report at least one incident, corresponding to a raw rate of 16.2%.

2.6 Methodological Considerations

Several characteristics shape the interpretation of the data:

- **Survey routing:** missing values in digital blocks may be structural rather than ordinary non-response.
- **Self-reporting:** access, use, knowledge, security behaviour, and incidents may be affected by recall or interpretation.
- **Pseudo-weighting:** population weights are used, but complete strata and cluster information is not incorporated.
- **Composite outcomes:** “at least one” indicators are interpretable but conceal breadth and type of activity.

Table 2: Direction correction used to construct the security-awareness index.

Item	Security-relevant content	Aware response	Recoding
QK7_4	A digital contract requires a paper signature	False	$1 - x$
QK7_5	Public data may be used for targeted offers	True	x
QK7_6	Cryptocurrency has legal-tender status	False	$1 - x$
QS2_6	Shares passwords with friends	Never	No reversal
QS2_7	Checks whether the provider is regulated	Always	$6 - x$
QS2_8	Shares financial information publicly	Never	No reversal
QS3_13	Regularly changes passwords	Completely	$6 - x$
QS4_1	It is safe to shop using public Wi-Fi	Disagree	No reversal
QS4_2	Website security matters before a transaction	Agree	$6 - x$
QS4_3	Terms and conditions are not important to read	Disagree	No reversal

- **Sparse cells:** some age–education or intense-use subgroups have small samples and should not drive substantive conclusions.
- **Incident ambiguity:** a reported outcome combines exposure with recognition, recall, and reporting behaviour.

3 Methodology

3.1 Overall Analytical Logic

The empirical workflow follows the complete pathway:

1. estimate overall Internet access and identify demographic concentration of exclusion;
2. estimate a weighted multivariable access model and adjusted probabilities;
3. use corrected Cramer’s V as an exploratory screening tool;
4. construct online-adoption and frequent-use indicators among connected adults;
5. compare age, education, item-level activities, and age–education interactions;
6. estimate a parsimonious weighted adoption model;
7. estimate the RQ2 fraud-exposure model;
8. describe incident rates by usage intensity and awareness;
9. estimate the RQ3 awareness, intensity, and interaction models;
10. integrate access, participation, exposure, and protection into a single interpretation.

The design prioritises interpretable descriptive statistics and regression models rather than a broad comparison of algorithms.

3.2 Weighted Descriptive Estimation

For a binary outcome y_i and positive weight w_i , the weighted prevalence is:

$$\hat{p}_w = \frac{\sum_i w_i y_i}{\sum_i w_i}. \quad (2)$$

The access analysis accompanies group percentages with an approximate Wilson interval using Kish’s effective

sample size [2]:

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \quad (3)$$

This is more appropriate than treating the weighted sum as a simple-random-sample count, but it does not replace a full survey-design variance estimator.

3.3 Corrected Cramer’s V

A bias-corrected Cramer’s V matrix is calculated for categorical variables in the access dataset. The measure is bounded between 0 and 1 and is used descriptively to identify the strongest unadjusted associations with Internet access. It is not interpreted as a causal effect and is not used as an automatic feature-selection algorithm.

3.4 Weighted Logistic Model for Internet Exclusion

The main access model uses `no_internet` as the dependent variable:

$$\begin{aligned} \text{logit}\{P(Y_i = 1)\} = & \beta_0 + \beta_A \text{Age}_i + \beta_E \text{Education}_i \\ & + \beta_G \text{Gender}_i + \beta_W \text{Work}_i \\ & + \beta_R \text{Region}_i + \beta_U \text{Urbanisation}_i. \end{aligned} \quad (4)$$

The model is estimated as a weighted binomial GLM with HC1 robust covariance. The reference categories are ages 18–29, man, university education, employed worker, North-West, and large city. Global Wald tests assess the joint significance of each categorical factor. Robustness checks compare weighted, unweighted, and weighted-plus-income specifications.

3.5 Average Adjusted Predictions for Access

For each age or education category, that category is assigned to every observation while all other characteristics remain at their observed values. Predicted probabilities are averaged with survey weights. Confidence intervals are generated from 2,000 simulated coefficient vectors after numerical correction of the covariance matrix to a positive semi-definite form.

3.6 Weighted Logistic Model for Online Adoption

Among connected respondents with a valid QP8 outcome, the adoption model is:

$$\text{logit}\{P(A_i = 1)\} = \alpha_0 + \alpha_A \text{Age}_i + \alpha_E \text{Education}_i. \quad (5)$$

The reference categories are ages 40–49 and upper-secondary education. Weights are normalised to mean one, and HC1 robust covariance is used. Adjusted adoption probabilities are reported as point estimates because the notebook does not provide uncertainty intervals for them.

3.7 Weighted Logistic Model for RQ2

Let F_i denote reported fraud exposure. The supplied RQ2 model is:

$$\text{logit}\{P(F_i = 1)\} = \gamma_0 + \gamma_1 U_i + \gamma_2 K_i + \gamma_3 T_i + \gamma_A \text{Age}_i + \gamma_E \text{Education}_i, \quad (6)$$

where U_i is Internet-use frequency, K_i is perceived digital/financial knowledge, and T_i is the number of financial tools. The reference groups are ages 18–39 and High education. The model reports coefficients, odds ratios, 95% confidence intervals, and p -values with HC1 robust standard errors.

The model estimates conditional associations. Non-significant age and education coefficients do not prove full mediation by use or knowledge. A formal mediation analysis would require an explicit indirect-effect model, stronger temporal assumptions, and adequate power.

3.8 Weighted Logistic Models for RQ3

RQ3 estimates three logistic regressions with `financial_incident` as the dependent variable.

Model 1 contains security awareness S_i only:

$$\text{logit}\{P(Y_i = 1)\} = \delta_0 + \delta_1 S_i. \quad (7)$$

Model 2 adds digital-finance intensity I_i :

$$\text{logit}\{P(Y_i = 1)\} = \delta_0 + \delta_1 S_i + \delta_2 I_i. \quad (8)$$

Model 3 adds the interaction and socio-demographic controls $\mathbf{X}_i$:

$$\text{logit}\{P(Y_i = 1)\} = \delta_0 + \delta_1 S_i + \delta_2 I_i + \delta_3 (S_i \times I_i) + \theta' \mathbf{X}_i. \quad (9)$$

The controls are age, education, gender, employment status, macro-region, and urbanisation. The final complete-case regression sample contains 3,686 respondents, including 650 who report at least one incident.

Because intensity is observed on a 1–4 scale, the awareness main effect in the interaction model refers to intensity zero and is not substantively interpreted. The relevant quantity is the combined awareness effect $\delta_1 + I\delta_3$ evaluated at representative intensity levels.

4 Results

4.1 RQ1a: Internet Access

4.1.1 Overall Access and Demographic Concentration

Among respondents with a valid response, 4,427 report Internet access and 415 report no access. After weighting, 89.42% have access and 10.58% remain excluded. The national average conceals a highly concentrated divide, as summarised in Table 3.

The exclusion rate is below 2% for adults younger than 50, rises to 5.11% at ages 50–59, reaches 19.28% at ages 60–69, and exceeds 42% among those aged 70 or more. Education shows an even steeper monotonic gradient, from 1.15% among university graduates to 62.02% among respondents with primary education or less.

4.1.2 Exploratory Association Screening

The corrected Cramer’s V matrix is consistent with the descriptive patterns. Education has the strongest association with Internet access ($V = 0.437$), followed by age group ($V = 0.404$) and employment status ($V = 0.365$). Income is weaker ($V = 0.133$), while urbanisation ($V = 0.059$), gender ($V = 0.039$), and macro-region ($V = 0.031$) show smaller unadjusted associations.

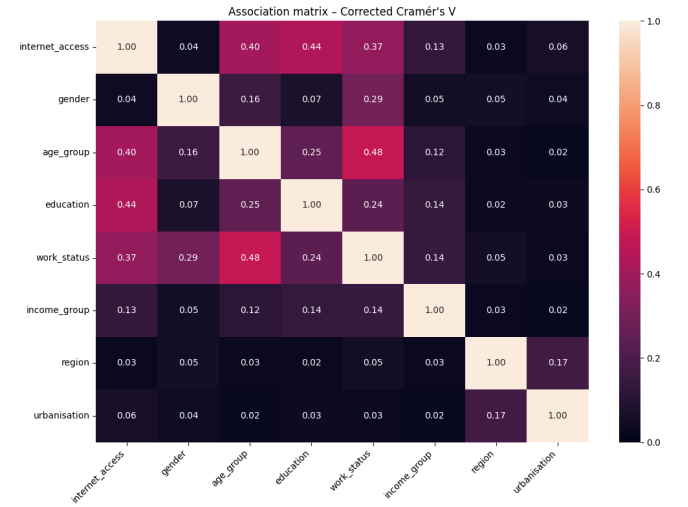


Figure 2: Corrected Cramer’s V association matrix. The matrix provides an exploratory rationale for prioritising age and education in the connected-sample analysis.

4.1.3 Intersection of Age and Education

The age–education heatmap shows that the two principal barriers reinforce one another. Among respondents aged 60–69, exclusion is 7.09% for university graduates, 9.38% for upper-secondary graduates, 31.41% for lower-secondary graduates, and 59.60% for those with primary education or less. Among people aged 70 or more, the corresponding rates reach 22.90%, 52.02%, and 69.60% for the available lower education groups. Cells with fewer

Table 3: Weighted percentage without Internet access.

Group	Weighted %	95% CI
<i>Age</i>		
18–29	1.34	0.56–3.15
30–39	0.70	0.29–1.67
40–49	1.86	1.10–3.15
50–59	5.11	3.75–6.94
60–69	19.28	16.26–22.70
70+	42.08	38.01–46.27
<i>Education</i>		
University	1.15	0.67–1.95
Upper secondary	4.89	4.03–5.91
Lower secondary	26.21	22.61–30.17
Primary or less	62.02	52.17–70.97

than 30 observations are flagged and should not be interpreted substantively.

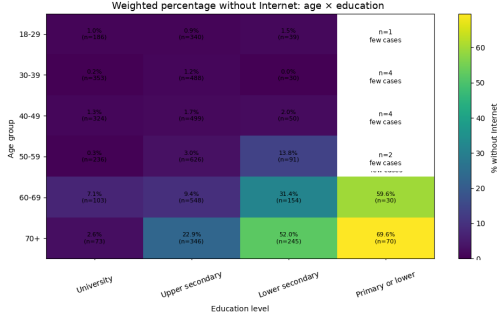


Figure 3: Weighted percentage without Internet access by age and education.

4.1.4 Multivariable Access Model

The weighted logistic model confirms that age and education remain strongly associated with exclusion after adjustment for gender, employment, macro-region, and urbanisation.

Table 4: Significant factors in the weighted logistic model for no Internet access.

Contrast	OR	95% CI	<i>p</i>	<i>PFDR</i>
Age 50–59 vs 18–29	3.06	1.15–8.12	0.02472	0.06180
Age 60–69 vs 18–29	5.81	2.17–15.58	0.00047	0.00196
Age 70+ vs 18–29	11.46	4.18–31.41	< 0.00001	0.00001
Upper secondary vs university	2.26	1.23–4.15	0.00875	0.02429
Lower secondary vs university	7.67	4.24–13.88	< 0.00001	< 0.00001
Primary or less vs university	26.07	12.93–52.56	< 0.00001	< 0.00001
Housework vs employed	2.63	1.71–4.06	0.00001	0.00006
Pensioner vs employed	2.94	1.90–4.54	< 0.00001	0.00001
Rural area vs large city	2.83	1.45–5.52	0.00232	0.00724
North-East vs North-West	0.55	0.39–0.78	0.00066	0.00234

Relative to ages 18–29, the odds of exclusion are approximately 5.81 times higher at ages 60–69 and 11.46 times higher at ages 70 or more. Relative to university education, the odds are 7.67 times higher for lower-secondary education and 26.07 times higher for primary education or less. Pensioners, people engaged in housework, and rural residents also show higher odds. Gender is not statistically significant in the main model.

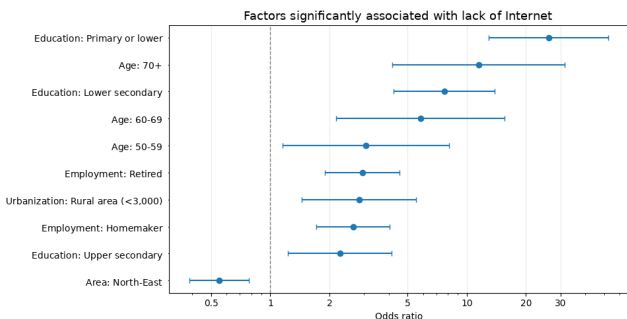
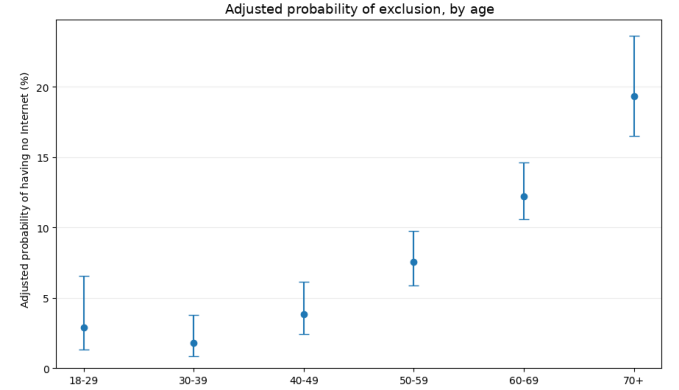


Figure 4: Odds ratios for statistically significant factors associated with Internet exclusion. The horizontal axis is logarithmic.

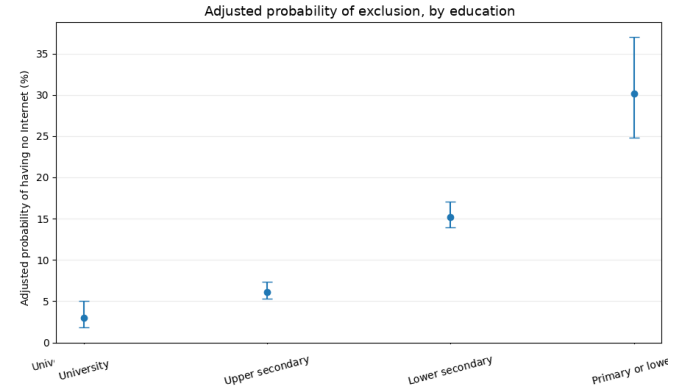
Global Wald tests indicate joint significance for age ($\chi^2 = 76.98$, $p < 0.001$), education ($\chi^2 = 189.17$, $p < 0.001$), employment status ($\chi^2 = 30.23$, $p = 0.00019$), macro-region ($\chi^2 = 13.53$, $p = 0.00894$), and urbanisation ($\chi^2 =$

18.72, $p = 0.00089$). Gender is not jointly significant ($p = 0.119$).

Adjusted exclusion probabilities rise from 2.90% at ages 18–29 to 19.31% at ages 70 or more. Across education, they rise from 3.01% for university graduates to 30.13% for primary education or less. The raw percentages are higher in some subgroups because adjusted predictions average over the observed distribution of the remaining covariates.



(a) Adjusted probability by age.



(b) Adjusted probability by education.

Figure 5: Average adjusted probability of having no Internet access, holding the distribution of the other covariates constant.

4.1.5 Robustness

Unweighted, weighted, and weighted-plus-income specifications produce directionally stable results. For example, the unweighted odds ratios for ages 60–69 and 70+ are 5.73 and 11.84, close to the weighted estimates of 5.81 and 11.46. Older age, lower education, and rural residence remain associated with exclusion across specifications.

4.2 RQ1b: Online Adoption and Frequency of Use

4.2.1 Transition from Connectivity to Adoption

Among connected respondents with a valid QP8 outcome, 1,400 report at least one product opened or subscribed entirely online and 2,849 report none. The weighted adoption rate is 31.08%. The count distribution is concentrated at zero: 2,938 respondents have no positive QP8 item, 722 have one, 450 have two, 143 have three, 68 have four, and 17 report all five; 89 cases have no usable count. Current activity is considerably broader. Among valid QP9 cases, 3,307 report at least one activity performed

Often or Very often, compared with 944 who do not. The weighted frequent-use prevalence is approximately 75%, and the mean QP9 score is 2.07 among respondents with at least four valid items.

Online adoption follows an inverted-U age pattern, peaking at 41.09% among ages 40–49. Frequent QP9 activity is higher than online onboarding in every age group and peaks at 87.65% among ages 30–39. Education displays a monotonic gradient for both outcomes.

Table 5: Weighted adoption and frequent activity among connected adults.

Group	QP8 %	QP9 frequent %
<i>Age</i>		
18–29	20.50	69.91
30–39	31.97	87.65
40–49	41.09	82.96
50–59	36.56	76.70
60–69	26.60	64.38
70+	19.25	49.09
<i>Education</i>		
Lower secondary or less	15.74	49.04
Upper secondary	31.48	77.24
University	41.43	88.81

4.2.2 Products Opened Entirely Online

Opening an account and requesting a card are the most common QP8 activities. Among ages 40–49, 21.0% report online account opening, 22.8% card requests, and 18.6% insurance. Credit is less common, peaking at 7.0%, and investment-platform onboarding remains rare in all age groups, with a maximum of 3.3%.

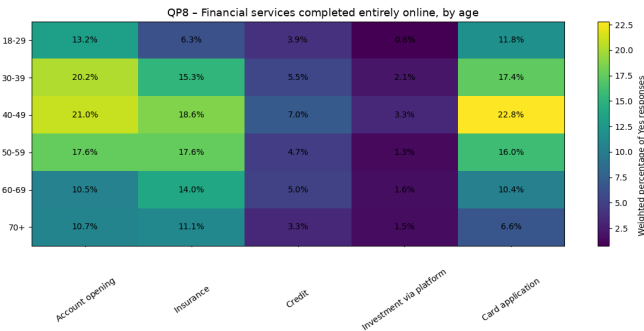


Figure 6: Weighted share completing each financial service entirely online by age.

As it is shown in Figure 6 education differentiates all activities. Among university graduates, 25.2% report requesting a card entirely online, 21.6% insurance, and 21.3% account opening. The corresponding rates among respondents with lower-secondary education or less are 7.9%, 5.3%, and 8.9%. Investment-platform adoption remains low even among graduates (2.6%).

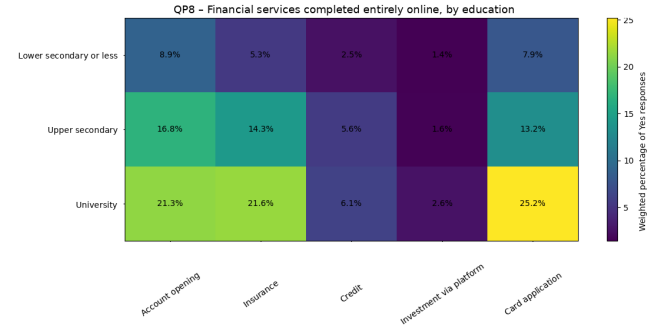


Figure 7: Weighted share completing each financial service entirely online by education.

4.2.3 Digital Activities Used Frequently

Balance and transaction checks are the most frequent QP9 activity, followed by bill payments and online purchases. Money transfers and smartphone payments occupy an intermediate position, while financial-product management and robo-advisor use are less common.

Among ages 30–39, 78.4% frequently check balances or transactions, 64.7% frequently pay bills, and 55.4% frequently make online purchases. The 40–49 group has the highest smartphone-payment rate (31.4%) and robo-advisor rate (21.9%). Among respondents aged 70 or more, the rates fall to 34.2%, 31.5%, and 19.5% for the three most common activities.

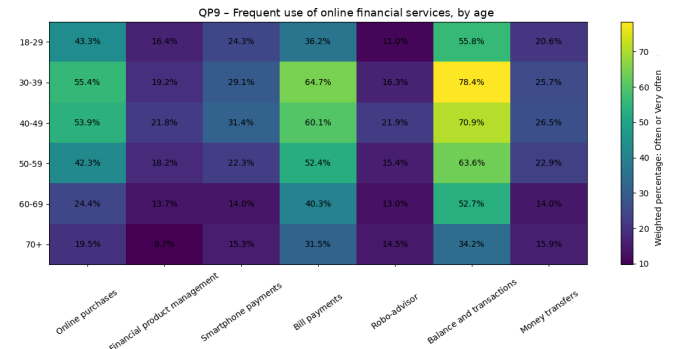


Figure 8: Weighted share reporting each digital financial activity Often or Very often by age

University graduates report frequent balance checks at 79.9%, bill payments at 62.8%, and online purchases at 55.8%. Among lower-secondary education or less, the corresponding rates are 36.4%, 32.1%, and 20.9%.

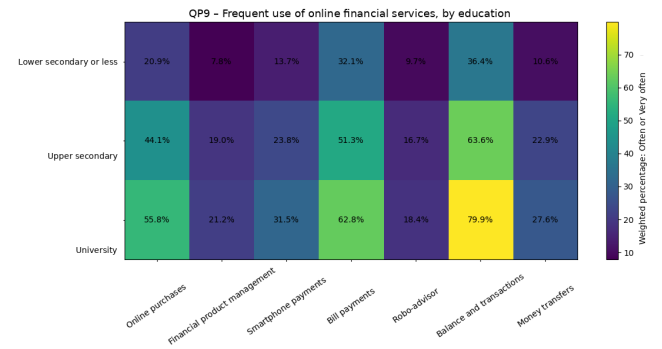


Figure 9: Weighted share reporting each digital financial activity Often or Very often by education

4.2.4 Age–Education Interaction

Online adoption is highest among university graduates aged 40–49 (51.07%), followed by university graduates aged 50–59 (48.93%) and 60–69 (42.40%). The lowest stable estimate is 9.01% among respondents aged 70 or more with lower-secondary education or less.

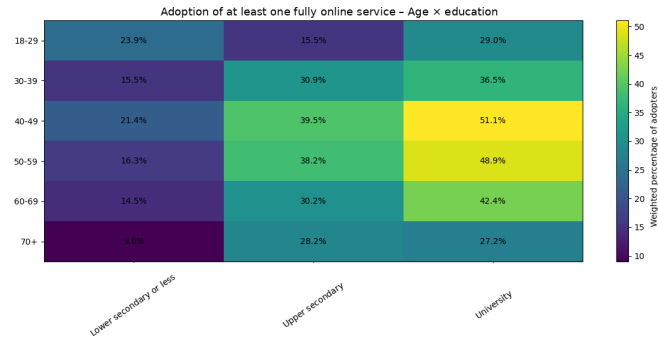


Figure 10: Weighted age–education interactions for online onboarding among connected adults.

Frequent use reaches 93.88% among university graduates aged 30–39, 92.86% among university graduates aged 40–49, and 90.05% among university graduates aged 50–59. It falls to 30.79% among the oldest, least educated group. Education remains strongly differentiating within age groups, while ageing remains consequential within education levels.

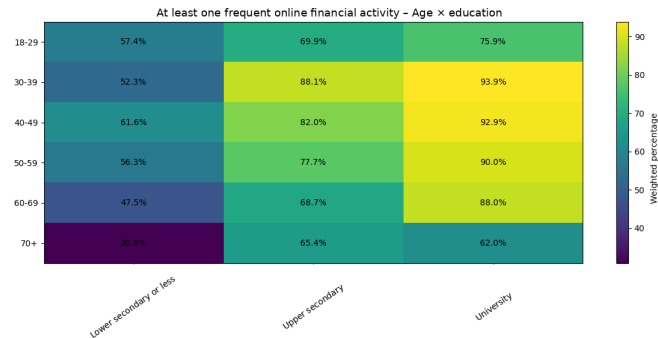


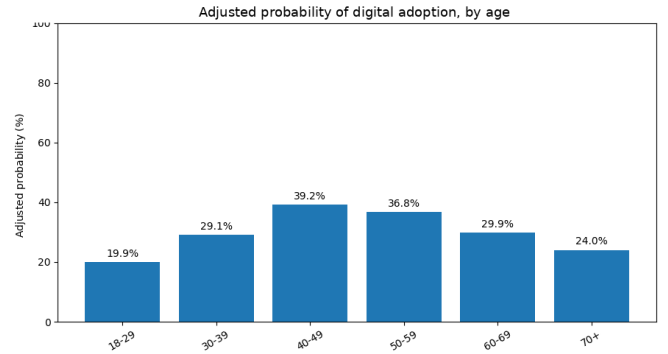
Figure 11: Weighted age–education interactions for frequent use among connected adults.

4.2.5 Weighted Adoption Model

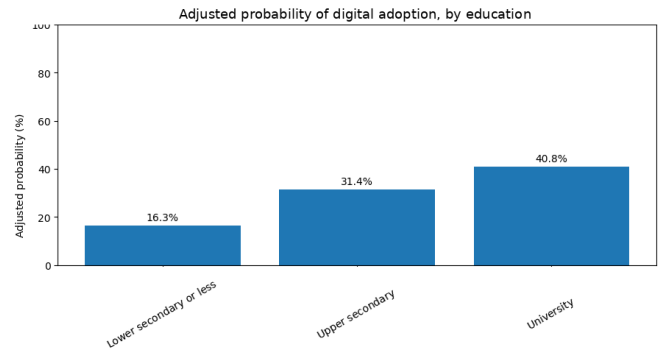
Relative to ages 40–49, the odds of adoption are lower among ages 18–29, 30–39, 60–69, and 70+. Ages 50–59 are not significantly different. Relative to upper-secondary education, lower-secondary education or less is associated with lower odds, while university education is associated with higher odds.

The Cox–Snell pseudo- R^2 is 0.0565, showing that age and education are relevant but explain only a limited share of individual differences. Adjusted adoption probabilities are 19.88%, 29.07%, 39.17%, 36.84%, 29.89%, and 24.04% across the six age groups. By education, they are 16.34%, 31.43%, and 40.79%.

The complete odds ratios, confidence intervals, and significance levels are reported in Table ??.



(a) Adjusted adoption by age.



(b) Adjusted adoption by education.

Figure 12: Average adjusted probabilities from the weighted adoption model. Point estimates are shown without confidence intervals.

4.3 RQ2: Activity and Reported Fraud Exposure

4.3.1 Regression Results

Table 7 reports the weighted binomial logistic model supplied for RQ2. The operational variables are statistically significant, while the age and education contrasts are not significant after adjustment.

4.3.2 Exposure through Digital Activity

A one-unit increase in Internet-use frequency is associated with 2.186 times the odds of reporting a fraud attempt or incident, corresponding to an estimated 118.6% increase in the odds. The pattern is consistent with an exposure mechanism: frequent use of digital banking and e-commerce creates more occasions for contact with fraudulent messages, websites, payment requests, and transactions. The coefficient must be interpreted on the engineered scale used in the supplied analysis.

4.3.3 Knowledge, Recognition, and Reporting

The largest coefficient concerns perceived digital/financial knowledge ($OR = 15.174$), with a wide confidence interval from 2.898 to 79.461. The supplied analysis interprets this as a possible awareness or reporting effect. More knowledgeable respondents may be better able to identify and classify suspicious events, whereas lower-knowledge users may fail to recognise or report the same experience. The estimate therefore does not demonstrate that knowledge increases underlying victimisation. Its magnitude and uncertainty require caution.

Table 6: Weighted logistic model for at least one QP8 online-adoption activity.

Contrast	OR	95% CI	<i>p</i>	Interpretation
Age 18–29 vs 40–49	0.374	0.294–0.477	< 0.001	Lower adoption
Age 30–39 vs 40–49	0.626	0.512–0.767	< 0.001	Lower adoption
Age 50–59 vs 40–49	0.903	0.742–1.099	0.307	No significant difference
Age 60–69 vs 40–49	0.653	0.517–0.824	< 0.001	Lower adoption
Age 70+ vs 40–49	0.480	0.357–0.645	< 0.001	Lower adoption
Lower secondary or less vs upper secondary	0.419	0.337–0.521	< 0.001	Lower adoption
University vs upper secondary	1.518	1.304–1.768	< 0.001	Higher adoption

Table 7: Weighted logistic model for reported digital financial fraud exposure. HC1 robust standard errors.

Predictor	Coef.	OR	95% CI	<i>p</i>	Interpretation
Intercept	−4.648	0.010	0.001–0.108	< 0.001	Reference-profile odds
Education: Medium vs High	0.220	1.246	0.587–2.646	0.566	Not significant
Education: Low vs High	−0.494	0.610	0.190–1.961	0.407	Not significant
Age: 40–59 vs 18–39	0.645	1.906	0.481–7.543	0.358	Not significant
Age: 60+ vs 18–39	0.172	1.187	0.366–3.849	0.775	Not significant
Internet-use frequency	0.782	2.186	1.249–3.826	0.006	Positive association
Digital/financial knowledge	2.720	15.174	2.898–79.461	0.001	Large positive association
Number of financial tools	−0.158	0.854	0.765–0.952	0.004	Negative association

4.3.4 Financial Tools and Demographic Controls

Each additional financial tool is associated with 0.854 times the odds of the reported outcome, equivalent to an estimated 14.6% reduction per unit. Possible explanations include more structured financial management, safer institutional relationships, product selection, income, or unobserved user characteristics. The model cannot distinguish among these mechanisms.

Age and education are not statistically significant after adjustment. This means that the model does not detect conditional demographic differences in the supplied specification; it does not prove that use and knowledge fully mediate age or education effects.

4.3.5 Answer to RQ2

RQ2 is supported with important interpretive qualifications. More intensive digital activity is positively associated with reported financial-fraud exposure, consistent with a larger operational attack surface. Perceived knowledge also has a large positive coefficient, but this is plausibly influenced by recognition and reporting rather than greater underlying victimisation. The number of financial tools is negatively associated with the reported outcome. Age and education do not show statistically detectable conditional associations once these operational variables are included.

The results therefore suggest that fraud reports are more closely related to how respondents interact with digital financial channels and recognise suspicious events than to demographic characteristics alone. They do not justify causal claims or a formal conclusion of complete mediation.

4.4 RQ3: Security Awareness at the Same Level of Exposure

4.4.1 Descriptive Evidence

Incident rates increase with usage intensity and decline across security-awareness tertiles.

Table 8: Weighted incident rate by usage-intensity group.

Usage group	<i>n</i> valid	Incident rate
Non-user	946	8.9%
Low use	2,094	12.5%
Medium use	956	33.3%
Intense use	85	23.4%

The intense-use estimate is based on a small group and should be treated as descriptive. The lower rate than the medium group is not sufficient evidence of a genuinely non-monotonic relationship.

Table 9: Weighted incident rate by security-awareness tertile.

Awareness tertile	<i>n</i> valid	Incident rate
Low awareness	1,199	23.3%
Medium awareness	1,261	16.4%
High awareness	1,281	12.6%

The joint distribution is particularly informative. Among low-intensity users, the weighted incident rate falls from 18.5% in the low-awareness group to 8.8% in the high-awareness group. Among medium-intensity users, it falls from 40.3% to 24.2%. Awareness makes little difference among non-users, for whom exposure is limited.

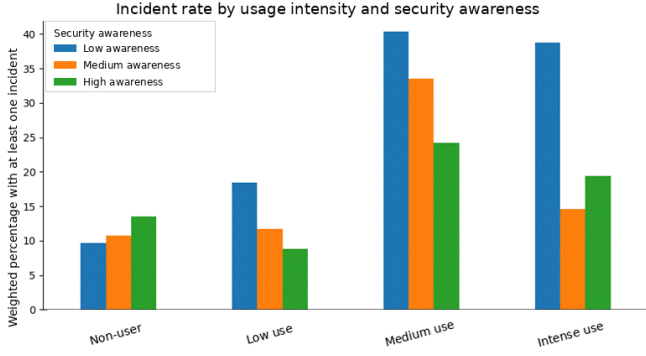


Figure 13: Weighted incident rate by usage-intensity group and security-awareness tertile. Cross-cells in the intense-use group are small and should be interpreted cautiously.

4.4.2 Regression Models

As summarised in Table 10, Model 1 (see characteristics in section 3.8) shows that a one-unit increase in the standardised awareness index is associated with an odds ratio of 0.42 ($p < 0.001$). In Model 2, the association remains strong after controlling for intensity (OR = 0.48, $p < 0.001$), while intensity is positively associated with incidents (OR = 2.30, $p < 0.001$).

In Model 3, the awareness–intensity interaction is negative and significant (OR = 0.62, $p = 0.006$). This means that the protective association of awareness becomes stronger as intensity rises. The isolated awareness OR of 1.41 is not substantively meaningful because it refers to intensity zero, outside the observed 1–4 scale.

Table 11: Approximate combined awareness effect at representative intensity levels.

Intensity	Log-odds	Approx. OR
Low (1.0)	−0.13	0.88
Below average (2.0)	−0.60	0.55
Above average (3.0)	−1.07	0.34
Intense (4.0)	−1.54	0.21

These are point estimates obtained by combining the awareness main effect and interaction coefficient. Formal confidence intervals require a delta-method or margins calculation accounting for coefficient covariance.

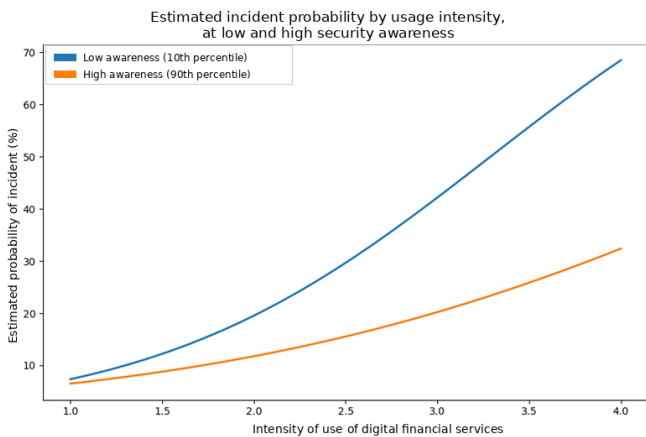


Figure 14: Predicted probability of a financial incident by usage intensity at the 10th and 90th percentiles of security awareness. The gap widens as intensity increases.

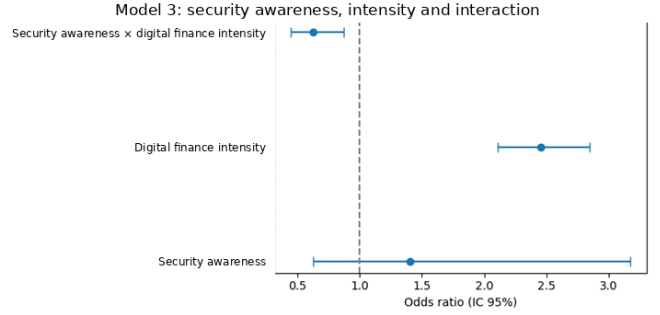


Figure 15: Odds ratios and 95% confidence intervals for security awareness, digital-finance intensity, and their interaction in Model 3.

4.4.3 Answer to RQ3

The evidence supports RQ3 with an important qualification. Greater security awareness is associated with a lower probability of reporting a financial incident after accounting for usage intensity, but the association is not constant across users. It is modest among people with little digital-finance activity and becomes progressively stronger among medium- and high-intensity users.

Security awareness therefore has its largest measurable protective value precisely where exposure is greatest. Read together with RQ2, the results provide a coherent interpretation: use intensity is a principal driver of exposure, while security awareness is a protective factor conditional on that exposure.

4.5 Integrated Interpretation of the Three Research Questions

The complete analysis identifies a sequence of barriers and risks rather than a single digital divide.

First, access remains the entry barrier. Internet exclusion is highly concentrated among older and less educated adults. These respondents cannot benefit from online financial services without assisted access or alternative channels.

Second, connectivity does not guarantee adoption. Although almost nine in ten adults have Internet access, only about three in ten connected adults report at least one product opened entirely online. Current digital activity is broader than online onboarding, demonstrating that acquisition channel and subsequent use are distinct dimensions.

Third, greater activity increases exposure. The RQ2 and RQ3 results consistently show that users who interact more intensively with digital financial channels report more incidents. This should not be interpreted as a reason to discourage digital participation; it means that the attack surface expands with the number and frequency of interactions.

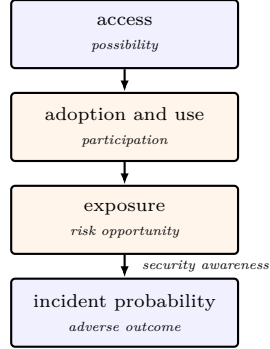
Fourth, awareness moderates exposure risk. Security awareness is associated with lower incident probability after intensity is considered, and its protective role is strongest among medium- and high-intensity users.

The final framework is summarised here below.

Table 10: Summary of weighted logistic models for financial incidents.

Model	<i>n</i>	Awareness OR	Intensity OR	Interaction OR	Pseudo- <i>R</i> ²
1. Awareness only	3,686	0.42***	–	–	0.019
2. + usage intensity	3,686	0.48***	2.30***	–	0.064
3. + interaction and controls	3,686	1.41 n.s.	2.45***	0.62**	0.089

Odds ratios are reported. *** $p < 0.001$; ** $p < 0.01$; n.s. = not significant at 5%. Model 3 controls for age, education, gender, employment status, macro-region, and urbanisation.

**Figure 16:** The final integrated framework: access, participation, exposure, and protection.

5 Conclusion

5.1 Final Answers to the Research Questions

RQ1a. Internet exclusion in Italy is not evenly distributed. It is concentrated among older adults, people with lower education, pensioners, people engaged in housework, and rural residents. Age and education remain strongly associated with exclusion after multivariable adjustment, while gender is not statistically significant in the main model.

RQ1b. Among connected adults, fully online onboarding remains incomplete. The weighted adoption rate is 31.08%, with the highest levels among middle-aged and university-educated respondents. Frequent use is substantially more widespread than fully online onboarding and is concentrated in routine activities such as balance checks, bill payments, and online purchases. The evidence therefore distinguishes access, acquisition channel, and current use.

RQ2. More intensive digital activity is positively associated with reporting a digital financial fraud attempt or incident. Perceived knowledge also has a large positive coefficient, plausibly reflecting recognition and reporting as well as underlying exposure. The number of financial tools is negatively associated with the reported outcome. Age and education are not statistically significant after adjustment, but this does not establish complete mediation.

RQ3. Security awareness is associated with lower incident probability after accounting for digital-finance intensity. The significant interaction shows that the protective association strengthens as intensity rises. Awareness has its largest measurable value among medium- and high-intensity users, the same groups for whom exposure is greatest.

Taken together, the results show that digital financial inclusion is not achieved merely by connecting individ-

uals to the Internet. Genuine inclusion requires access, usable and accessible services, effective participation, and protection proportionate to exposure.

5.2 Policy and Stakeholder Implications

The findings support differentiated interventions across the pathway.

- **For older and less educated adults without Internet:** maintain accessible non-digital channels, provide assisted digital access, and avoid online-only communication strategies.
- **For connected non-adopters:** simplify onboarding, reduce administrative complexity, improve accessibility, and provide guided support during first use.
- **For lower-education users:** begin with practical operations such as checking balances, paying bills, verifying providers, and recognising suspicious requests before promoting complex products.
- **For medium- and high-intensity users:** prioritise targeted security-awareness campaigns, in-app warnings, transaction-level prompts, periodic risk reminders, and rapid reporting tools.
- **For banks and FinTech firms:** distinguish product acquisition, post-acquisition use, exposure, recognition, and protection. A low QP8 rate does not imply inactivity, and a high reported-incident rate may partly reflect better detection.
- **For regulators and consumer associations:** evaluate attempted contact, successful victimisation, reporting, and financial loss as separate outcomes whenever possible.

The practical message is a shift from uniform digital-literacy campaigns to risk-based support. Security communication should be proportionate to users' level of exposure and embedded in the moments when potentially risky operations occur.

5.3 Strengths of the Study

- **Sequential design:** the analysis follows the pathway from access to adoption, activity, exposure, and protection.
- **Use of survey weights:** descriptive estimates are population-oriented rather than based only on raw sample proportions.
- **Interpretable modelling:** odds ratios, confidence intervals, global tests, and adjusted probabilities directly address the research questions.
- **Exploratory screening:** corrected Cramer's *V* provides a transparent rationale for prioritising age and

education.

- **Detailed behavioural decomposition:** QP8 and QP9 remain available at activity level rather than being reduced only to composite outcomes.
- **Interaction analysis:** RQ3 moves beyond an average awareness effect and shows that protection depends on exposure intensity.

5.4 Limitations

- **Cross-sectional design:** temporal order and causal direction cannot be established. An incident may itself increase subsequent awareness.
- **Self-reported measures:** access, activity, knowledge, security behaviour, and incidents may be affected by recall and social-desirability bias.
- **Incomplete survey design:** weights are used, but complete strata and primary sampling unit information is not incorporated.
- **Outcome ambiguity:** incident indicators may combine attempted exposure, successful victimisation, recognition, and reporting.
- **RQ2 reproducibility:** the supplied summary does not report the final sample size or every coding threshold for engineered predictors.
- **Wide uncertainty for perceived knowledge:** the RQ2 confidence interval is very broad, so the OR of 15.174 should not be read as precise.
- **Composite awareness index:** the ten items combine knowledge, behaviour, and attitudes. Internal consistency and dimensionality should be formally tested.
- **Small intense-use cells:** estimates for intense users and awareness cross-cells are descriptive because some groups contain fewer than 30 observations.
- **Approximate combined effects:** RQ3 odds ratios at representative intensity levels are point estimates without delta-method confidence intervals.
- **Different category definitions:** age and education are grouped differently across components; the final repository should document and justify these choices.

5.5 Directions for Further Research

1. **Complete RQ2 documentation.** Report the exact QP10 aggregation, predictor ranges, missing-data rules, final sample size, and weight normalisation.
2. **Validate the security-awareness index.** Assess internal consistency, item-total relationships, and dimensionality. If the items measure multiple constructs, separate knowledge, behaviour, and attitude subscales.
3. **Estimate formal marginal effects.** Use a delta-method or margins procedure to obtain confidence intervals for the awareness effect at each intensity level.
4. **Separate incident processes.** Model attempted scams, phishing, unauthorised card use, unrecognised transactions, successful victimisation, reporting, and financial loss as distinct outcomes.
5. **Add adoption mechanisms.** Include general digital familiarity, perceived usefulness, trust, income, product

ownership, and channel preference in hierarchical adoption models.

6. Construct mutually exclusive user profiles. Cross-classify Internet access, onboarding, intensity, incident history, and awareness to identify digitally excluded adults, connected non-adopters, active low-risk users, and highly exposed users with limited awareness.

7. Use longitudinal or repeated cross-sectional data. This would help distinguish whether awareness precedes lower incident probability or is partly a consequence of prior adverse experiences.

8. Apply full survey-design estimation and external validation. Incorporate strata and primary sampling units when available, and compare survey findings with administrative fraud reports or later survey waves.

5.6 Final Takeaway

The central contribution of the project is to show that digital financial inclusion and digital financial security are inseparable. Access is necessary but not sufficient. Adoption creates participation, participation creates exposure, and security awareness can reduce the probability that exposure becomes an adverse incident. The strongest policy response is therefore a combined strategy: remove access barriers, make digital services easier to adopt and use, and concentrate security support where digital-finance intensity – and consequently potential exposure – is highest.

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